

CDAC- Cloud Computing

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Question 1: Docker Installation & Nginx

Deployment

Task

1. Install Docker on your cloud virtual machine.
2. Verify Docker installation.
3. Download the Nginx image.
4. Run the Nginx container.
5. Access the Nginx welcome page from your browser.

Required Screenshots

- Docker version output
- Docker images output
- Docker ps output showing Nginx running
- Browser showing "Welcome to nginx!"

```
[[ec2-user@ip-172-31-36-162 ~]]$ docker --version
Docker version 25.0.14, build 0bab007
[[ec2-user@ip-172-31-36-162 ~]]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx          latest    936fef290c8f   2 days ago    161MB
myimage        latest    623ac033f11b   7 weeks ago   100MB
ubuntu         latest    30ba44506a6d   7 weeks ago   100MB
hello-world    latest    e2ac70e7319a   2 months ago  10.1kB
[[ec2-user@ip-172-31-36-162 ~]]$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[[ec2-user@ip-172-31-36-162 ~]]$ docker run -d -p 80:80 nginx
3334f1f5a95ce3aed4ddd4c7f66796371f89cd24f46c44e86fa292bf022f7125
[[ec2-user@ip-172-31-36-162 ~]]$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
3334f1f5a95c   nginx     "/docker-entrypoint...." 39 seconds ago Up 37 seconds  0.0.0.0:80->80/tcp, :::80->80/tcp  nifty_chandrasekhar
```

Question 2: Docker Volume Practical

Task

1. Create a Docker volume named: studentdata
2. Run an Ubuntu container using the volume.
3. Create a file: /data/info.txt
4. Write your name and roll number inside the file.
5. Display the contents of the file.

Required Screenshots

- Volume creation
- Running container
- File creation
- File contents displayed using cat

```
[ec2-user@ip-172-31-36-162 ~]$ docker volume create studentdata
studentdata
[ec2-user@ip-172-31-36-162 ~]$ docker volume ls
DRIVER      VOLUME NAME
local       mydata
local       studentdata
[ec2-user@ip-172-31-36-162 ~]$ docker run -it -v mydata:/data ubuntu bash
root@e61872597386:/# exit
exit
[ec2-user@ip-172-31-36-162 ~]$ docker run -it -v studentdata:/data ubuntu bash
root@83dbae444a8b:/# echo "Meet Mahesh Alshi" > /data/info.txt
root@83dbae444a8b:/# echo "Roll No. 24101B0049" >> /data/info.txt
root@83dbae444a8b:/# cat /data/info.txt
Meet Mahesh Alshi
Roll No. 24101B0049
root@83dbae444a8b:/# █
```

Question 3: Build Your First Docker Image

Task

Create a custom Docker image that displays the following message when executed:

Cloud Computing Practical Exam Completed Successfully

Build the image and run it successfully.

Required Screenshots

- Dockerfile
- docker build output
- docker images output
- docker run output

Hint

Relevant Dockerfile keywords:

- FROM
- CMD
- COPY

Think about:

- Which base image should be used?

Ans: The base image depends on the application requirements. For a general Linux environment, ubuntu is commonly used, while lightweight applications may use alpine.

- How does Docker know what command to execute when the container starts?

Ans: Docker executes the command specified in the CMD or ENTRYPOINT instruction of the Dockerfile when the container starts.

- When would COPY be useful while building an image?

Ans: COPY is used to transfer files from the host system into the Docker image, such as application code, configuration files, or web pages needed by the container.

```
[ec2-user@ip-172-31-36-162 ~]$ docker run -it -v studentdata:/data ubuntu bash
[root@83dbae444a8b:/# echo "Meet Mahesh Alshi" > /data/info.txt
[root@83dbae444a8b:/# echo "Roll No. 24101B0049" >> /data/info.txt
[root@83dbae444a8b:/# cat /data/info.txt
Meet Mahesh Alshi
Roll No. 24101B0049
[root@83dbae444a8b:/# exit
exit
[ec2-user@ip-172-31-36-162 ~]$ mkdir practical
[ec2-user@ip-172-31-36-162 ~]$ cd practical
[ec2-user@ip-172-31-36-162 practical]$ nano Dockerfile
[ec2-user@ip-172-31-36-162 practical]$ cat Dockerfile
FROM ubuntu
CMD L "echo", "Cloud Computing Practical Exam Completed Successfully"]
[ec2-user@ip-172-31-36-162 practical]$ docker build -t myimage .
[+] Building 0.1s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 179B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/ubuntu:latest
=> exporting to image
=> => exporting layers
=> => writing image sha256:54e17e06c5a0bf35de7e1914e8391029e6b7fd0107171ff47e68eb5c91752010
=> => naming to docker.io/library/myimage
[ec2-user@ip-172-31-36-162 practical]$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    936fef290c8f   2 days ago    161MB
myimage       latest    54e17e06c5a0   7 weeks ago    100MB
<none>        <none>    623ac033f11b   7 weeks ago    100MB
ubuntu        latest    30ba44506a6d   7 weeks ago    100MB
hello-world   latest    e2ac70e7319a   2 months ago   10.1kB
```

```

[[ec2-user@ip-172-31-36-162 practical]$ docker run practical
/bin/sh: 1: L: not found
[[ec2-user@ip-172-31-36-162 practical]$ nano Dockerfile
[[ec2-user@ip-172-31-36-162 practical]$ cat Dockerfile
FROM ubuntu
CMD [ "echo", "Cloud Computing Practical Exam Completed Successfully"]
[[ec2-user@ip-172-31-36-162 practical]$ docker build -t myimage .
[+] Building 0.1s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 179B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/ubuntu:latest
=> exporting to image
=> => exporting layers
=> => writing image sha256:654ae01180258ca34bc640616e7217a2f4baf2800dc6b507a894aa72db5ca122
=> => naming to docker.io/library/myimage
[[ec2-user@ip-172-31-36-162 practical]$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
nginx               latest             936fef290c8f       2 days ago         161MB
practical           latest             54e17e06c5a0       7 weeks ago        100MB
myimage             latest             654ae0118025       7 weeks ago        100MB
<none>              <none>             623ac033f11b       7 weeks ago        100MB
ubuntu              latest             30ba44506a6d       7 weeks ago        100MB
hello-world         latest             e2ac70e7319a       2 months ago       10.1kB
[[ec2-user@ip-172-31-36-162 practical]$ docker run myimage
Cloud Computing Practical Exam Completed Successfully
[[ec2-user@ip-172-31-36-162 practical]$

```

Bonus Question

Explain in 2–3 lines each:

1. Difference between a Docker Image and a Docker Container.

Ans: A Docker Image is a read-only template that contains the application and its dependencies. A Docker Container is a running instance of that image where the application actually executes.

2. Why are Docker Volumes used?

Ans: Docker Volumes are used to store data permanently outside the container. This ensures that data is not lost even if the container is stopped, removed, or recreated.